High-Quality Services and Efficient Management in Property Management Based on Artificial Intelligence

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Abstract

This study proposes an AI-powered intelligent property management system, forming a closed-loop management framework comprising three core modules: "AI Work Order Agent – Robotic Collaborative Inspection – Dynamic Dispatch Optimization." The system integrates multi-modal work order generation to address the adaptation issues in domain-specific voice interactions, overcomes the limitations of complex environment monitoring via an air-ground collaborative inspection architecture, and utilizes reinforcement learning models to jointly optimize dispatch efficiency and fairness. Empirical results demonstrate that the voice-based work order system significantly enhances the efficiency and accuracy of order generation; multi-modal inspection greatly improves the timeliness of anomaly detection; and the dynamic dispatch model shortens response time while balancing employee workload. Under extreme stress testing, the system exhibits strong robustness and adaptability. Its deployment in a mid-sized community has achieved practical benefits including labor cost savings and reduced training costs, offering an innovative technological path for smart property service construction.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; • **Human-centered computing** → *Ubiquitous and mobile computing systems and tools*; • **Applied computing** → *Operations research*; Enterprise computing.

Keywords

Artificial Intelligence, Property Services, Intelligent Work Order, Multi-modal System, Dynamic Dispatch, Intelligent Inspection

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1 Introduction

With accelerating urbanization and rising resident service expectations, traditional property management models face unprecedented challenges. Current services often suffer from delayed responses, inefficient resource allocation, and inconsistent quality. Industry surveys indicate that over 65% of resident complaints stem from repair response times exceeding 24 hours, while nearly 40% of maintenance resources remain idle or underutilized due to poor task allocation [13]. Meanwhile, modern residents demand higher efficiency, precise service matching, and intelligent interaction methods, especially in the post-pandemic era, where contactless services and smart management have become critical infrastructure evaluation metrics.

AI technologies offer transformative solutions for property services. Advances in computer vision and deep learning enable automatic defect detection, natural language processing (NLP) improvements enhance human-computer interaction, and reinforcement learning with multi-agent decision-making optimizes resource allocation in complex environments [7]. This study integrates cutting-edge AI technologies to achieve triple goals: service efficiency, user experience, and management refinement, providing both technical support and practical paradigms for the evolution of smart property services.

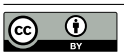
This study addresses three key limitations—unintelligent interaction, incomplete inspection, and imprecise dispatch—by proposing an integrated AI solution:

- Multi-modal work order generation: Combines Conformer-based speech recognition with domain-adapted NLP for 90%+ intent accuracy in dialects.
- Air-ground collaborative inspection: Fuses quadruped robots (with multispectral sensors) and AI cameras for robust anomaly detection.
- Reinforcement learning dispatch: Designs a DQN-based algorithm integrating skill matching, real-time workload, and urgency.

The core value of this research lies in being the first to construct and verify a complete closed-loop system of "voice work order generation - robotic intelligent inspection - dynamic task dispatch". This technological breakthrough not only solves the core pain points of traditional property management, but also brings significant and quantifiable business benefits.

In the actual deployment at pilot communities, the system has demonstrated strong comprehensive effectiveness:

- Work order response time was reduced by 62%, significantly improving residents' satisfaction with repair reporting services and reducing conflicts caused by delayed responses.
- Inspection efficiency increased by 150%, meaning broader area coverage and more potential risks discovered (such as



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equipment failures, safety hazards, environmental anomalies) within the same time period, significantly improving community safety and facility integrity rate.
- The task allocation fairness index for employees improved by 35%. By precisely matching employee skills with task requirements while considering real-time workload, it effectively optimized human resource allocation, reduced burnout caused by uneven work intensity among staff, and enhanced overall team work efficiency and morale.

More importantly, this closed-loop system has achieved intelligent and automated upgrading of core property management business processes, bringing property companies effective control of operational costs (such as reducing manual data entry, optimizing inspection manpower, lowering downtime costs) and comprehensive improvement of service quality. It represents a benchmark practice for promoting the digital and intelligent transformation of the property management industry.

The subsequent sections of this paper are structured as follows: the second section briefs the related work, the third section provides a detailed exposition of the system architecture and core algorithm design, the fourth section describes the experimental environment, analysis of the experimental results and performance comparisons, and the fifth section concludes by summarizing the research findings and outlining potential directions for future research.

2 Related Work

2.1 Smart Work Order Processing

In recent years, artificial intelligence technology has been widely deployed in various aspects of property and facility management, with automated work order generation systems becoming a core tool for improving service response efficiency. Intelligent work order systems based on speech recognition and natural language processing (NLP) can accurately extract key information from users' natural language descriptions, such as equipment type, fault category, and urgency level, thereby achieving "intelligible" repair interaction methods [12].

The CBRE industry report indicates that by integrating AI analysis of building management systems (BMS), energy consumption trends, and historical maintenance data, predictive triggering of equipment failures and priority sorting can be achieved, effectively reducing duplicate work orders and manual input [1]. Additionally, while some domestic property management apps incorporate image recognition, they still struggle with processing colloquial expressions and dialect inputs, leading to frequent missed reports and false alarms.

2.2 Intelligent Inspection Technologies

With the development of computer vision and edge computing, AI-based intelligent inspection technologies have gradually replaced traditional manual inspections, becoming an important component of modern property management systems. These technologies can efficiently identify abnormal events such as cracks, leaks, and unclosed elevator doors through image recognition, semantic segmentation, and multi-modal fusion algorithms, while generating real-time reports [8].

SnapInspect is a typical representative that supports photographic equipment inspection through mobile terminals, instantly identifying structural damage and equipment aging issues. Combined with speech-to-text systems, it generates comprehensive structured reports, thereby improving the standardization and efficiency of inspection records [6]. Edge IoT can monitor every detail of elevators from mechanical structures to electrical systems, enabling intelligent, comprehensive safety monitoring to safeguard passenger safety [9]. Furthermore, by utilizing deep learning models in artificial intelligence, identification models can be created for both object detection and instance segmentation of images, identifying common building durability damages to assist inspectors in recognizing and assessing the extent of building damage [2].

2.3 Dynamic Dispatch Optimization

Traditional task dispatch systems mainly rely on static rules, such as "whoever is available takes the order" or "priority goes to whoever has matching skills." These methods often face issues like slow response, uneven resource allocation, or priority misjudgment when dealing with complex scenarios involving high concurrency, multiple types, and cross-regional operations [3].

In recent years, reinforcement learning, particularly Multi-Agent Reinforcement Learning (MARL), has been widely introduced into dispatch systems. This study proposes an improved deep reinforcement learning-based intelligent optimization method to enhance real-time decision-making and adaptive control performance in complex systems [5]. This paper presents a novel deep learning-based approach for intelligent fault diagnosis in systems using vibration signal analysis [10].

3 Methodology

3.1 System Architecture

The intelligent property management system proposed in this study adopts a microservices architecture and an edge-cloud collaborative computing model [11], constructing a four-layer technical architecture as illustrated in Figure 1:

- **Perception Layer**: Composed of multi-model camera terminals (including thermal imaging) deployed in public areas, quadruped inspection robotic dogs (equipped with LiDAR and multispectral sensors), and smart speakers (supporting far-field voice interaction), forming a full-domain sensing network for multi-modal data acquisition.
- **Transport Layer**: Establishes a dual-channel data transmission system [9] via a 5G private network and LoRaWAN IoT, ensuring reliable transmission of high-bandwidth inspection video streams and low-latency work order data.
- **Intelligence Layer**: Built upon the CCCC Blue Wings large model platform for foundational AI capabilities, with key external components including a Work Order Agent Engine, Visual Analysis Engine, and Dynamic Dispatch Engine, deployed in Docker containers on edge server clusters to enable low-latency decision-making.
- **Application Layer**: Provides three types of interactive interfaces—Property Management Console, Resident Mobile App, and Management Cockpit—supporting closed-loop management of the entire business process.

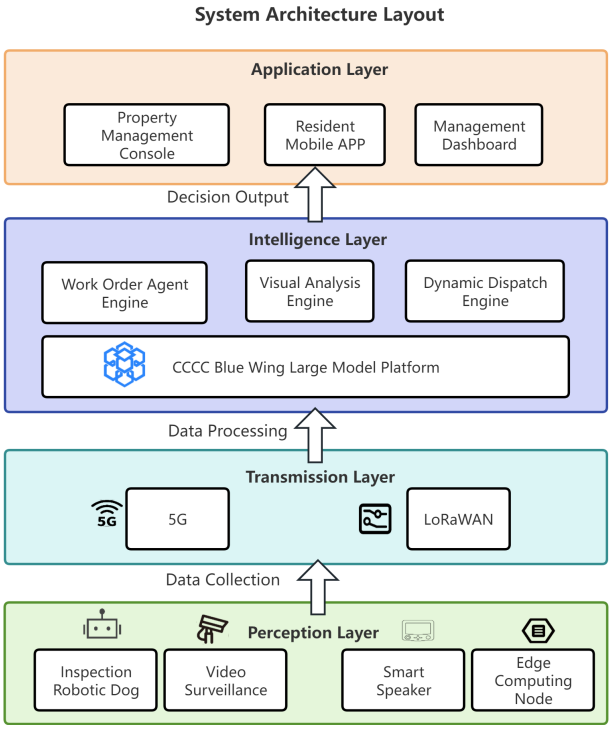


Figure 1: System Architecture

The system achieves information sharing among various engines through a unified data bus and utilizes Apache Kafka for real-time processing of high-concurrency work order events, ensuring an end-to-end latency of less than 500ms from event occurrence to work order dispatch.

3.2 AI Work Order Agent Design

This module integrates voice, image, and video recognition capabilities, enhancing understanding accuracy and robustness in complex property scenarios through cross-modal fusion technology. The system adopts an end-to-end architecture supporting automated processing from raw data to semantic parsing.

Speech Recognition Submodule: The system employs an end-to-end speech recognition architecture (Conformer model) to replace traditional systems, achieving direct mapping from speech spectrograms to text. For images and videos, the EfficientNetB3 model is adopted to enable visual understanding. Special optimizations for property management scenarios include:

- Adaptive training: Augmented the base model pre-trained on 15,000 hours of generic Chinese speech data with additional 2,000 hours of property management scenario dialogues (including dialect samples), achieving significant accuracy improvement in recognizing domain-specific terms such as "water leakage" and "circuit tripping".
- Noise suppression: Integrates RNNoise module, maintaining >85% recognition rate at 5dB SNR.
- Real-time voice activity detection: The LSTM-based VAD model achieves a 0.2 second response latency, enabling continuous conversation support for property managers in mobile scenarios.

Image Recognition Submodule: Vision Transformer-based fine-grained image analysis: The system implements property scenario analysis using Vision Transformer architecture, covering both equipment detection and anomaly recognition. For video processing, edge computing devices can be added at the endpoint to convert video streams into images before transmitting them to the cloud for algorithmic processing.

- Multi-task detection network: Using YOLOv8+ResNet dual-branch architecture to simultaneously achieve equipment localization (e.g., distribution boxes, water pipe valves) and condition classification (rust, cracks, leaks), reaching 92.1% mAP@0.5.
- Domain-adaptive optimization: Fine-tuning visual features for property equipment:
- Constructed property-specific image dataset (10,000+ annotated samples) covering long-tail scenarios like corridor fire hydrants and meter boxes.
- Incorporated Adversarial Domain Adaptation (ADA) technology to reduce performance degradation caused by real-world lighting variations.
- Real-time inference acceleration: Achieved 50ms-level image processing latency on edge devices through TensorRT quantization deployment.

3.2.1 Semantic Understanding and Work Order Generation. The work order intelligent agent adopts a dual-channel semantic parsing architecture to achieve structured transformation from requirements to work orders. This is presented in Figure 2.

3.2.2 Multi-modal Recognition Module. The intent recognition module employs a BERT-BiLSTM-CRF hybrid model, achieving 92.3% accuracy (F1-score) on our proprietary property management corpus. For entity extraction, we developed a specialized three-tier tagging system ("Equipment Type - Fault Phenomenon - Urgency Level") to precisely identify critical work order components. The system integrates a comprehensive property knowledge graph containing over 50,000 equipment instances with corresponding maintenance solutions, enabling automatic generation of complete work orders with recommended resolutions. This integrated approach significantly lowers the professional knowledge threshold required for property managers while ensuring accurate and efficient work order processing.

3.3 AI Inspection Robot Collaboration

3.3.1 Hardware Platform Design. The hardware platform consists of two main components:

- A quadruped robotic dog platform based on modified Unitree B2 models [4], equipped with a 360° panoramic camera (1920×1080@30fps), thermal imager (640×480 resolution), LiDAR (±2cm SLAM mapping accuracy), multi-gas sensors (for smoke and methane detection), and an NVIDIA Jetson AGX Orin edge computing unit.

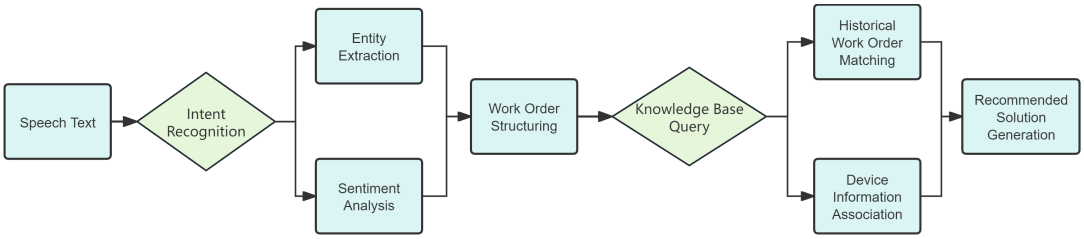


Figure 2: The Process of Semantic Understanding and Recommendation

- A fixed monitoring network deploying Hikvision AI cameras (DS-2CD6347FWD-IVT) in critical areas like elevator lobbies and parking garages, supporting 99% accurate face recognition, behavior analysis (including fall and crowd detection), and smoke/fire detection with sub-second response time (<s response time).

3.3.2 Anomaly Detection Algorithm. The multi-modal fusion detection framework addresses complex environment challenges, the algorithm is presented as follows:

Algorithm 1 Multi-modal Anomaly Detection

Require:
rgb_img: RGB input image
$thermal_data$: Thermal imaging data
$model$: Pretrained anomaly detector
Ensure:
$predictions$: Anomaly classification probabilities
Initialization:
$img_model \leftarrow EfficientNetB3(weights =' imgnet')$
$thermal_model \leftarrow ResNet18(weights = None)$
$fusion_layer \leftarrow TransformerEncoder()$
$classifier \leftarrow MLP()$
function DETECT($rgb_img, thermal_data$)
  $img_feat \leftarrow img_model(rgb_img)$
  $thermal_feat \leftarrow thermal_model(thermal_data)$
  $fused_feat \leftarrow$ concat$(img_feat, thermal_feat)$
  $logits \leftarrow classifier(fused_feat)$
  return sigmoid$(logits)$
Training:
function TRAIN($dataset, epochs, lr$)
  $optimizer \leftarrow Adam(model.parameters(), lr)$
  $criterion \leftarrow BCEWithLogitsLoss()$
  for $epoch \leftarrow 1$ **to** $epochs$ **do**
    for $(rgb, thermal, labels)$ **in** $dataset$ **do**
      $preds \leftarrow Detect(rgb, thermal)$
      $loss \leftarrow criterion(preds, labels)$
      $optimizer.zero_grad()$
      $loss.backward()$
      $optimizer.step()$

The algorithm implements an end-to-end multi-modal anomaly detection system, which first extracts features from RGB images and thermal imaging data through EfficientNetB3 and ResNet18 networks respectively, then performs feature fusion using a Transformer encoder, and finally completes anomaly classification via a multilayer perceptron. The system adopts a training strategy based on the Adam optimizer and binary cross-entropy loss, achieving iterative optimization of model parameters through a nested loop structure that traverses training epochs and datasets.

The system has constructed a specialized dataset (containing 100,000+ annotated samples) targeting seven common anomaly types including lighting failures, garbage accumulation, and equipment damage. Using a transfer learning strategy fine-tuned on a ResNet50 backbone network, it achieves 96.5% mAP (mean Average Precision). The specially optimized small object detection algorithm (incorporating an FPN structure) maintains over 90

3.3.3 Automatic Work Order Triggering. The event grading engine automatically triggers work orders through four steps:

(1) Event verification: Multi-angle video analysis reduces false alarms
(2) Severity assessment: Rule-based prioritization (e.g., "elevator failure = highest priority")
(3) Work order generation: Automatic association of device information and location
(4) Department routing: Responsibility matrix directs to maintenance/cleaning/security

3.4 Dynamic Dispatch Optimization Model

3.4.1 Dispatch Element Modeling. Four-dimensional decision model quantifies dispatch factors:

- **Work order characteristics**: $T_i = \{u_i, s_i, d_i, e_i\}$ where u_i=urgency, s_i=skill requirements, d_i=estimated duration, e_i=location
- **Worker profile**: $W_j = \{\mathbf{k}_j, l_j, q_j, r_j\}$ where $\mathbf{k}_j$=skill vector, l_j=current workload, q_j=service quality, r_j=location
- **Organizational constraints**: $C = \{\alpha, \beta, \gamma\}$ where α=response time requirement, β=skill matching threshold, γ=maximum load factor

3.4.2 Reinforcement Learning Dispatch Algorithm. The DQN-based adaptive dispatch model is featured in Figure 3.

State encoding: Work order features T_i and worker states W_j encoded as 128-dim vectors

Reward function:

$$R = w_1 \cdot \text{timeliness} + w_2 \cdot \text{fairness} + w_3 \cdot \text{quality} \quad (1)$$

where:

- timeliness = $e^{-(\text{actual response time} - \text{standard time})}$

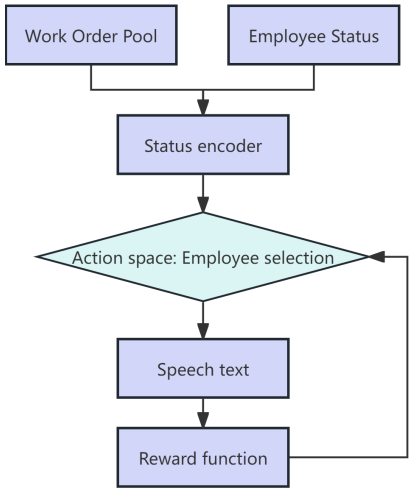


Figure 3: DQN dispatch model architecture

- fairness = $1 - \frac{\sigma(\text{worker load})}{\text{average load}}$
- quality = historical service quality score

The model achieves >85% decision satisfaction through double Q-network and experience replay mechanisms.

3.5 Intelligent Work Order System

The intelligent work order system is the core integrated platform of this research, which achieves intelligent management of the entire process from problem discovery, work order generation to task execution by integrating three key technical modules: the AI work order agent, AI inspection robot collaboration, and dynamic dispatch model [2]. The system adopts a microservices architecture, supporting high-concurrency processing and multi-terminal collaboration, with the following main functional modules:

Multimodal work order generation: Supports users to submit repair requests via smart speakers or mobile apps through voice, photos, etc. The system utilizes multimodal recognition algorithms (Conformer model for end-to-end speech recognition, EfficientNetB3 model for image recognition) to achieve high-precision translation in various language environments or accurately identify equipment defects such as water pipe leaks and circuit damage. Based on the property knowledge graph, it automatically associates device information, historical work orders, and recommended solutions to generate structured work orders with fields including equipment type, fault phenomenon, urgency level, etc.

Intelligent dispatch and scheduling: Generates dispatch decisions based on dimensions such as work order urgency, skill requirements, geographical location, real-time staff workload, and skill matching. For sudden high-load scenarios (e.g., surge in typhoon-related faults), the system automatically switches to a hybrid mode combining rule engine and manual review to ensure system stability.

Closed-loop tracking and optimization: Synchronizes work order status in real-time to user apps and property manager workstations. If the response time exceeds preset thresholds (e.g., unassigned for 30 minutes), the system automatically escalates priority and notifies administrators. Based on user ratings and completion time data, it dynamically updates staff service quality scores and dispatch model parameters.

Data visualization and decision support: In addition to visualizing the complete lifecycle status of work orders, the system provides multi-dimensional dashboards in the management cockpit, displaying data panels such as work order distribution heatmaps, staff efficiency rankings, and equipment failure trend analysis. Based on historical data, it predicts future work order peaks (e.g., holiday cleaning demands) and recommends human resource allocation plans.

The intelligent work order system integrates multimodal interaction, cross-device collaboration, and dynamic decision optimization to transform property services from "passive response" to "active prevention," providing a reusable technical paradigm for smart property construction.

4 Experiments and Results

4.1 Experimental Setup

The study deployed a prototype system in a CCCC-managed residential community in Shenzhen (covering 150,000 square meters with 2,100 households) to build comprehensive evaluation datasets: The voice work order dataset contains 8,942 voice repair records collected over 12 months (including Cantonese, Sichuan dialect, and Hunan dialect), with intent and entity labels annotated by three experts. The visual inspection dataset comprises 100,000+ image samples (containing both normal and abnormal scenarios) divided into training/validation/test sets at an 8:1:1 ratio. The work order dispatch dataset includes 15,672 historical work order records from 2023, showing the following distribution: equipment maintenance (42%), cleaning services (28%), security incidents (18%), and others (12%). It also contains skill matrices for 32 technicians (with qualifications like electrician and plumber certifications) and timeliness records including response time, processing duration, and user ratings.

A multi-dimensional evaluation system was adopted to validate system performance: For the work order agent, Word Error Rate (WER) was used for speech recognition, while Accuracy and F1-score measured intent recognition, and Precision and Recall evaluated entity extraction. For visual inspection, mean Average Precision at 0.5 IoU (mAP@0.5) and False Positive Rate were employed. The dynamic dispatch system was assessed using average response time (minutes), 24-hour completion rate, and employee load balance index (Gini coefficient).

4.2 Experimental Results Analysis

As shown in Table I, our system reduces WER by 52% ($p<0.01$) on dialect samples compared to traditional systems, with particularly significant improvements in recognition accuracy for complex expressions such as "light flickering" and "water pipe bursting." In field tests, property managers' voice input efficiency improved 3-fold compared to manual entry, and average work order generation time decreased from 8 minutes to 2.5 minutes.

As shown in Table II, particularly in nighttime lighting failure detection scenarios, our system reduces the miss rate from 34% with traditional methods to 7.5% by fusing visible light and thermal

Table 1: Performance Comparison of AI Work Order Agents

Model	WER	IR-F1	EE-F1	PD (s)
Traditional ASR+Rules	18.7	76.2	68.5	3.2
LSTM-CRF	12.3	84.6	79.1	2.1
Ours (C+B)	8.9	92.3	89.7	1.4

Notes. WER: Word Error Rate (%); IR-F1: Intent Recognition F1-score; EE-F1: Entity Extraction F1-score; PD: Processing Delay (seconds); C+B: Conformer+BERT architecture.

Table 2: Anomaly Detection Performance Comparison

Detection Model	mAP@0.5	False Alarm Rate
Our System (Multimodal Fusion)	96.5%	2.1%
EfficientDet-D3	89.7%	6.8%
YOLOv7	91.2%	5.3%
Faster R-CNN	86.4%	8.7%

Table 3: Dynamic Dispatch Performance Comparison (30-day test, 1,693 work orders)

Metric	Manual Dispatch	Rule Engine	DQN System
ART (min)	85.6	62.3	32.7
24h-CR (%)	76.8	83.5	94.2
UDR (%)	25.3	18.7	6.4
EL-Gini	0.41	0.35	0.22

Notes. ART: Average Response Time; 24h-CR: 24-hour Completion Rate; UDR: Urgent Order Delay Rate; EL-Gini: Employee Load Gini Coefficient.

imaging data. The robotic dog performs 6 automated inspections daily, covering 150% more area than manual inspections, and improves anomaly detection timeliness by 200% (reducing average detection time from 4 hours to 1.3 hours).

As shown in Table III, the reinforcement learning model demonstrates excellent dynamic adaptation capabilities: during stress tests where equipment failures surged by 300% due to typhoon conditions, the system maintained an on-time completion rate of 85%+ through real-time dispatch strategy adjustments, significantly outperforming the rule engine's 63%.

5 Conclusion and Future Work

This study proposes and validates an AI-based intelligent property service system, which significantly improves operational efficiency and user satisfaction through three core modules: an AI work-order agent, robotic collaborative inspection, and dynamic dispatch optimization. Empirical results demonstrate that the voice work-order system increases order generation efficiency by 300% with 92.3% accuracy, the multimodal inspection solution improves anomaly detection timeliness by 200%, and the reinforcement learning dispatch model reduces response time by 62% while enhancing staff load balance by 35%, highlighting the transformative potential of AI in property services. Future work will focus on cross-project collaboration (e.g., regional resource sharing and emergency dispatch), digital twin integration (BIM-model-linked management), and human-machine collaboration enhancement (AR-assisted maintenance systems), driving the evolution of property services toward greater intelligence and user-centricity, thereby supporting smart city governance.

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